Vector autoregression I

This version: October 29, 2026

ECMA 31150, Spring 2026

University of Chicago

Department of Economics

Mikkel Plagborg-Moller

Plan for this slide deck

1. Analysis of multiple time series
   - Cross-correlogram
2. Autoregressive distributed lag model
3. Granger causality: which variables predict each other?
4. Vector autoregression

Multiple time series

- So far we've discussed how to model a single time series $\{y_t\}$.
- But in practice we have access to 1000s of macro time series.
  - There are more than 800k on FRED, and counting!
- We often want to study the relationship between two or more series.
- We want to forecast using several relevant predictors.

Cross-correlogram

- Take two (jointly) covariance stationary time series, y_t and x_t.
- Define the population **cross-correlogram**:

$$\rho_{y,x}(\ell) \equiv \text{Corr}(y_t, x_{t-\ell}).$$

  Measures how much lags (or leads) of x predict y.

  - Exercise: show that $\rho_{y,x}(-\ell) = \rho_{x,y}(\ell)$.

- Natural estimator is the sample cross-correlogram:

$$\hat{\rho}_{y,x}(\ell) \equiv \begin{cases} \frac{1}{T}\sum_{t=\ell+1}^{T}(y_t - \bar{y})(x_{t-\ell} - \bar{x})/(\hat{\sigma}_y\hat{\sigma}_x) & \text{for } \ell \geq 0, \\ \hat{\rho}_{x,y}(-\ell) & \text{for } \ell < 0, \end{cases}$$

  where $\hat{\sigma}_y, \hat{\sigma}_x$ are the sample standard deviations.

THE WALL STREET JOURNAL.

Wall Street's Favorite Recession Indicator Is in a Slump of Its Own

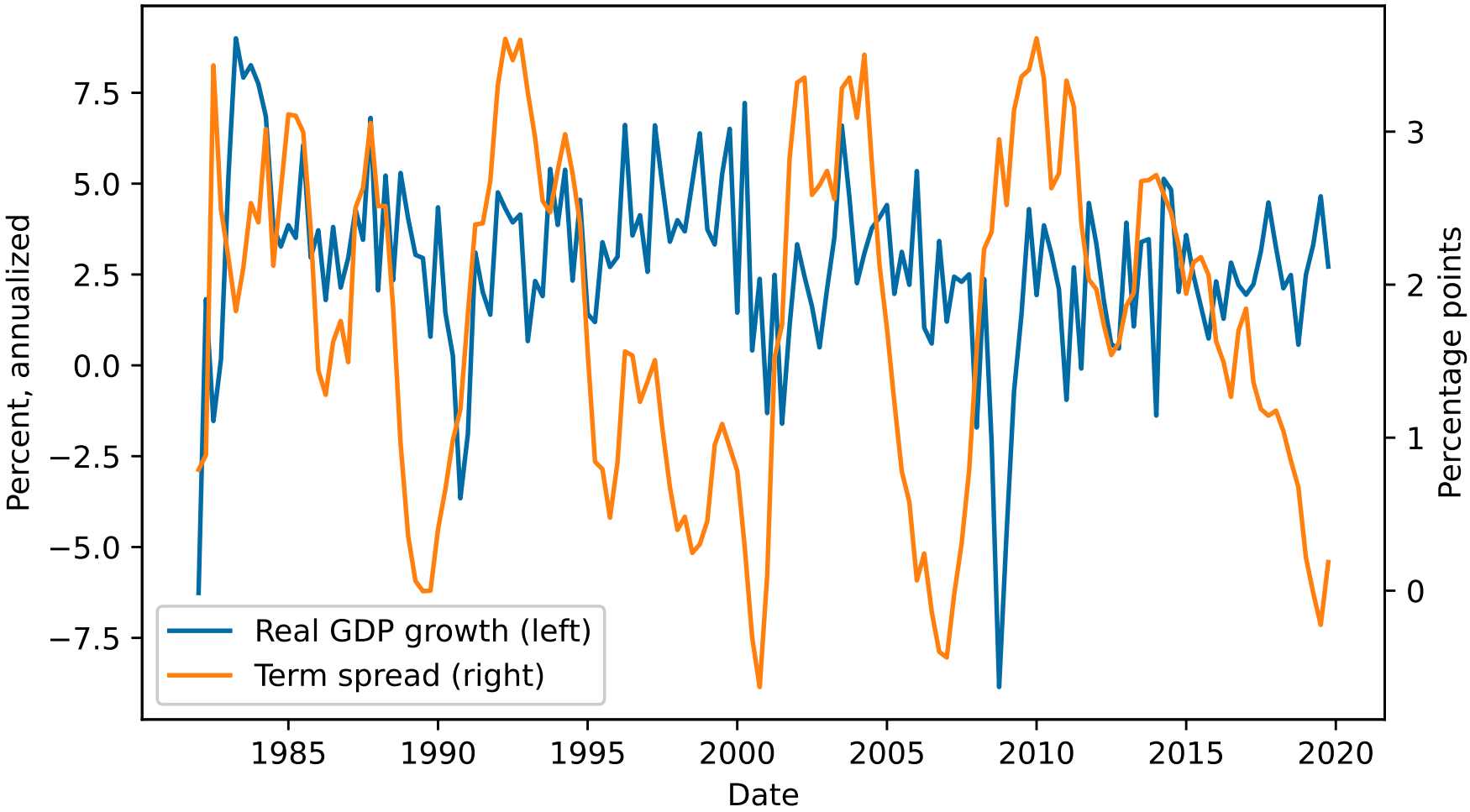
By Sam Goldfarb Follow *and* Peter Santilli Follow
May 28, 2024 5:30 am ET

One of Wall Street's favorite recession indicators looks broken.

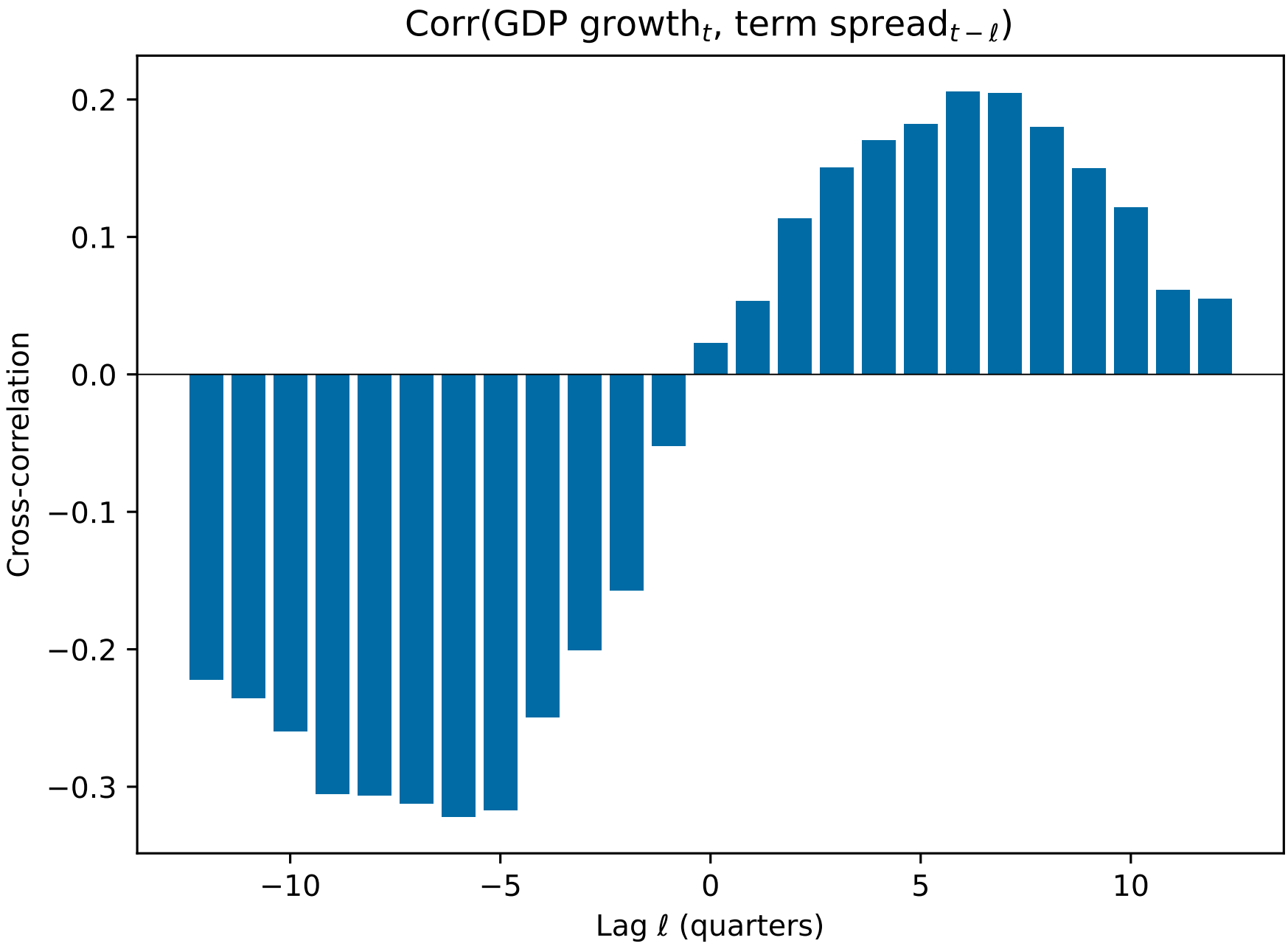
An anomaly known as an inverted yield curve, in which yields on short-term Treasurys exceed those of longer-term government debt, has long been taken as a nearly surefire signal that an economic pullback looms. In each of the previous eight U.S. downturns, that has happened before the economy sputtered. There haven't been any glaring false alarms.

Now, though, that streak is threatened. The yield curve has been inverted for a record stretch—around 400 trading sessions or more by some measures—with no signs of a major slowdown.

https://www.wsj.com/finance/investing/wall-streets-favorite-recession-indicator-is-in-a-slump-of-its-own-ee885cb9



Term spread = difference between yields on 10-year and 3-month U.S. Treasury bonds


Corr(GDP growth$_t$, term spread$_{t-\ell}$)
Cross-correlation
0.2
0.1
0.0
−0.1
−0.2
−0.3
−10
−5
0
5
10
Lag ℓ (quarters)

Autoregressive distributed lag model

- To use $\{x_t\}$ for forecasting $\{y_t\}$, we can add its lags as additional covariates in an AR specification:

$$y_t = c + a_1 y_{t-1} + \cdots + a_p y_{t-p} + b_1 x_{t-1} + \cdots + b_q x_{t-q} + u_t,$$

  where we assume

$$E[u_t \mid \{y_s, x_s\}_{s<t}] = 0.$$

- Called an **autoregressive distributed lag** (ADL) model.
- Optimal forecast:

$$\begin{aligned} y_{t|t-1} &\equiv E[y_t \mid \{y_s, x_s\}_{s<t}] \\ &= c + a_1 y_{t-1} + \cdots + a_p y_{t-p} + b_1 x_{t-1} + \cdots + b_q x_{t-q}. \end{aligned}$$

- Estimate parameters by OLS w. heteroskedasticity-robust SE.

```
n_lags = 2

# Generate lagged variables
reg_df = df[['growth', 'spread']].copy()
for l in range(1, n_lags + 1):
    reg_df[f'growth_lag{l}'] = reg_df['growth'].shift(l)
    reg_df[f'spread_lag{l}'] = reg_df['spread'].shift(l)

# OLS estimation
y = reg_df['growth']
X_cols = [f'growth_lag{l}' for l in range(1, n_lags + 1)] + \
    [f'spread_lag{l}' for l in range(1, n_lags + 1)]

X = sm.add_constant(reg_df[X_cols])

model = sm.OLS(y, X, missing='drop').fit(cov_type='HC1')
print(model.summary())
```

OLS Regression Results

Dep. Variable:	growth	R-squared:	0.260
Model:	OLS	Adj. R-squared:	0.239
Method:	Least Squares	F-statistic:	11.49
Date:	Mon, 06 Apr 2026	Prob (F-statistic):	3.94e-08
Time:	09:28:26	Log-Likelihood:	-322.65
No. Observations:	150	AIC:	655.3
Df Residuals:	145	BIC:	670.4
Df Model:	4		
Covariance Type:	HC1		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.9231	0.409	2.259	0.024	0.122	1.724
growth_lag1	0.3438	0.105	3.280	0.001	0.138	0.549
growth_lag2	0.2128	0.080	2.652	0.008	0.056	0.370
spread_lag1	-0.2585	0.304	-0.849	0.396	-0.855	0.338
spread_lag2	0.4460	0.329	1.355	0.175	-0.199	1.091

(rest of output omitted)

Granger causality

- We say that $\{x_t\}$ **Granger-causes** $\{y_t\}$ if the former provides any additional forecasting power over and above lags of the latter:

$$E[y_t \mid \{y_s, x_s\}_{s<t}] \neq E[y_t \mid \{y_s\}_{s<t}].$$

  - Confusing terminology: this is not about causal inference in the counterfactual (*ceteris paribus*) sense.
  - "The weather forecast Granger-causes the weather."

- In the ADL model

$$y_t = c + a_1 y_{t-1} + \cdots + a_p y_{t-p} + b_1 x_{t-1} + \cdots + b_q x_{t-q} + u_t,$$

$\{x_t\}$ Granger-causes $\{y_t\}$ if and only if $b_\ell \neq 0$ for at least one $\ell = 1, \ldots, q$.

  - Use F-test to test null hypothesis of no Granger causality.

```
spread_lags = [f'spread_lag{l}' for l in range(1, n_lags + 1)]
restriction = ' = '.join(spread_lags) + ' = 0'
f_test = model.f_test(restriction)
print(restriction)
print(f_test)
```

Result:

```
spread_lag1 = spread_lag2 = 0
<F test: F=1.2094836277525283, p=0.30134309292175676, df_denom=145, df_num=2>
```

Can't reject that the spread does *not* Granger-cause growth.

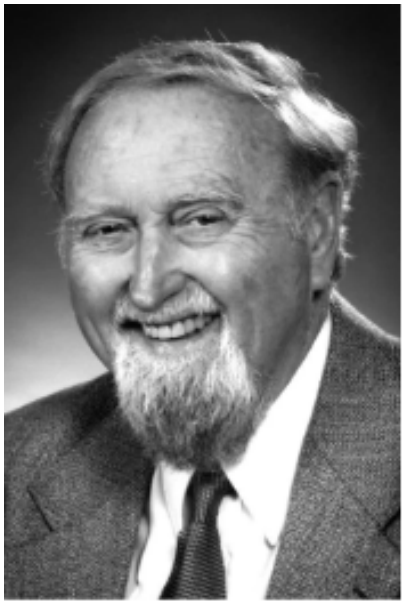


Photo from the Nobel Foundation archive.

Clive W.J. Granger
Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2003

Born: 4 September 1934, Swansea, United Kingdom

Died: 27 May 2009, San Diego, CA, USA

Affiliation at the time of the award: University of California, San Diego, CA, USA

Prize motivation: “for methods of analyzing economic time series with common trends (cointegration)”

https://www.nobelprize.org/prizes/economic-sciences/2003/granger/facts/

ADL model: multi-step forecast

- Consider ADL model with $p = q = 1$:

$$y_t = c + \rho y_{t-1} + b x_{t-1} + u_t.$$

- How to forecast two periods ahead?

$$\begin{aligned} y_{t+2} &= c + \rho y_{t+1} + b x_{t+1} + u_{t+2} \\ &= c + \rho(c + \rho y_t + b x_t + u_{t+1}) + b x_{t+1} + u_{t+2} \end{aligned}$$

$$\Downarrow$$

$$E[y_{t+2} \mid \{y_s, x_s\}_{s \le t}] = (1+\rho)c + \rho^2 y_t + \rho b x_t + b E[x_{t+1} \mid \{y_s, x_s\}_{s \le t}].$$

- But the model doesn't tell us what the last conditional expectation equals.
  $\Longrightarrow$ Solution: specify a second ADL model for x_t!

Vector autoregression

- ADL models for both y_t and x_t (new notation):

$$y_t = c_1 + A_{11} y_{t-1} + A_{12} x_{t-1} + u_{1t},$$
$$x_t = c_2 + A_{21} y_{t-1} + A_{22} x_{t-1} + u_{2t},$$

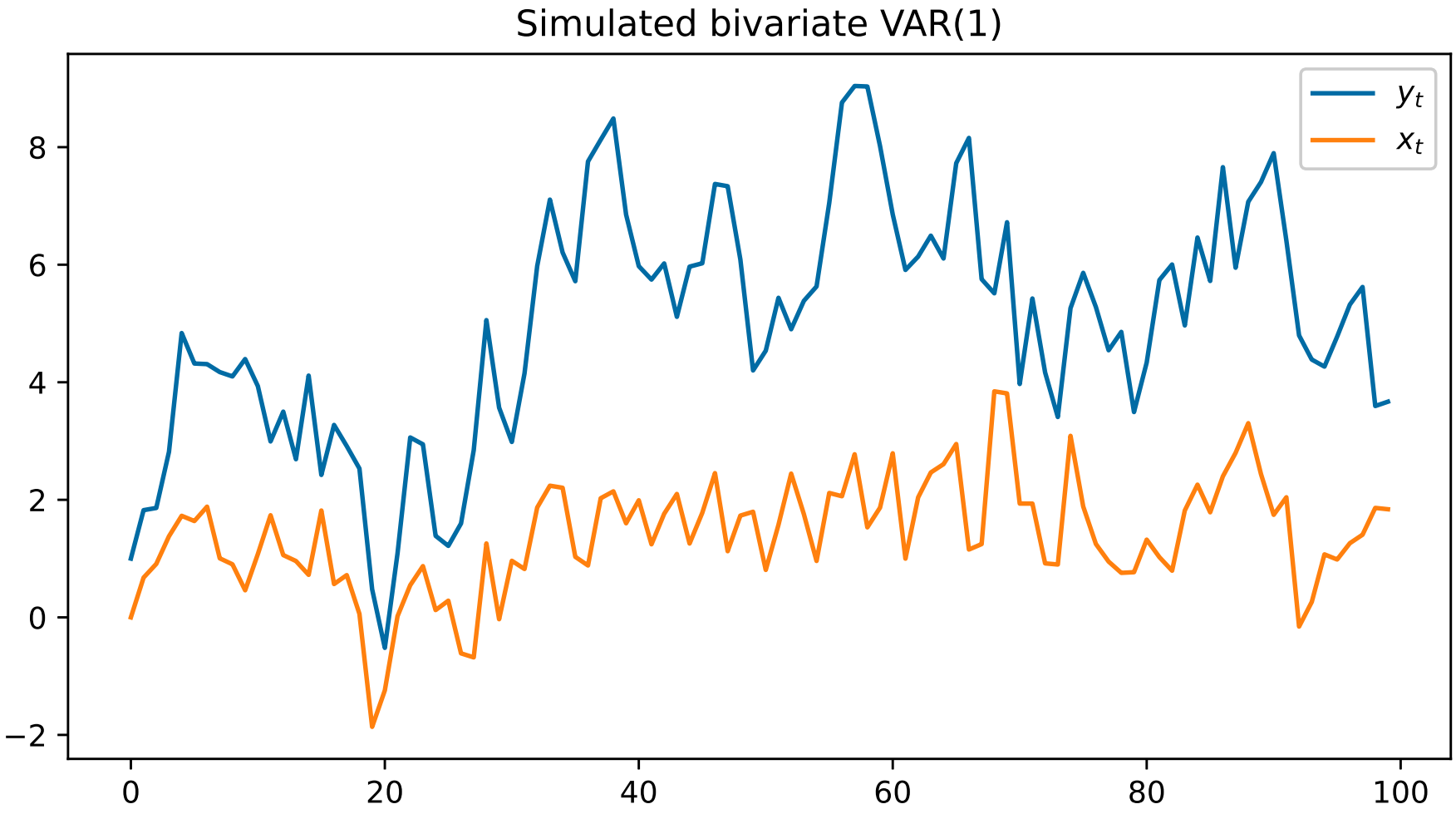
$$E[u_{jt} \mid \{y_s, x_s\}_{s<t}] = 0, \quad j = 1, 2.$$

- Convenient to write this with vectors and matrices:

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}\boldsymbol{y}_{t-1} + \boldsymbol{u}_t, \quad E[\boldsymbol{u}_t \mid \{\boldsymbol{y}_s\}_{s<t}] = \boldsymbol{0}, \quad \text{where}$$

$$\boldsymbol{y}_t \equiv \begin{pmatrix} y_t \\ x_t \end{pmatrix}, \quad \boldsymbol{c} \equiv \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}, \quad \boldsymbol{A} \equiv \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad \boldsymbol{u}_t \equiv \begin{pmatrix} u_{1t} \\ u_{2t} \end{pmatrix}.$$

- Called a **vector autoregression** of lag order 1, VAR(1).



$$\boldsymbol{y}_t = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0.8 & 0.1 \\ 0.2 & 0.3 \end{pmatrix} \boldsymbol{y}_{t-1} + \boldsymbol{u}_t, \quad \boldsymbol{u}_t \overset{iid}{\sim} N\left(\boldsymbol{0}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 0.5 \end{pmatrix}\right)$$

VAR(1): more variables

- With the vector notation, we can easily generalize our model to contain n variables:

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}\boldsymbol{y}_{t-1} + \boldsymbol{u}_t, \quad E[\boldsymbol{u}_t \mid \{\boldsymbol{y}_s\}_{s<t}] = \boldsymbol{0}, \quad \text{where}$$

$$\underbrace{\boldsymbol{y}_t \equiv \begin{pmatrix} y_{1t} \\ \vdots \\ y_{nt} \end{pmatrix}}_{n\times 1}, \quad \underbrace{\boldsymbol{c} \equiv \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}}_{n\times 1}, \quad \underbrace{\boldsymbol{A} \equiv \begin{pmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{n1} & \cdots & A_{nn} \end{pmatrix}}_{n\times n}, \quad \underbrace{\boldsymbol{u}_t \equiv \begin{pmatrix} u_{1t} \\ \vdots \\ u_{nt} \end{pmatrix}}_{n\times 1}.$$

- Henceforth I will be using the y_{it} notation, not y_t and x_t.

VAR(1): innovations

- The innovation process $\{\boldsymbol{u}_t\}$ is **multivariate white noise**:

$$\text{Corr}(u_{it}, u_{j,t-\ell}) = 0 \quad \text{for all } \ell \neq 0 \text{ and } i, j \in \{1, \ldots, n\},$$

  i.e., the cross-correlogram between $\{u_{it}\}$ and $\{u_{jt}\}$ is 0 at all leads and lags $\ell \neq 0$.

- We still allow the elements of $\boldsymbol{u}_t$ to be *contemporaneously* correlated:

$$\boldsymbol{\Sigma} \equiv \text{Var}(\boldsymbol{u}_t) = \begin{pmatrix} \text{Var}(u_{1t}) & \text{Cov}(u_{1t}, u_{2t}) & \cdots & \text{Cov}(u_{1t}, u_{nt}) \\ \text{Cov}(u_{2t}, u_{1t}) & \text{Var}(u_{2t}) & \cdots & \text{Cov}(u_{2t}, u_{nt}) \\ \vdots & & \ddots & \vdots \\ \text{Cov}(u_{nt}, u_{1t}) & \text{Cov}(u_{nt}, u_{2t}) & \cdots & \text{Var}(u_{nt}) \end{pmatrix}$$

  need not be a diagonal matrix.

VAR(1): moments

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}\boldsymbol{y}_{t-1} + \boldsymbol{u}_t$$

- Suppose $\{\boldsymbol{y}_t\}$ is covariance stationary. Then

$$E[\boldsymbol{y}_t] = \boldsymbol{c} + \boldsymbol{A}E[\boldsymbol{y}_t] \quad \Longrightarrow \quad \boxed{E[\boldsymbol{y}_t] = (\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{c}},$$

  where $\boldsymbol{I}$ is the $n \times n$ identity matrix.

- Moreover, we can solve for $\text{Var}(\boldsymbol{y}_t)$ through the "Lyapunov equations"

$$\boxed{\text{Var}(\boldsymbol{y}_t) = \boldsymbol{A}\,\text{Var}(\boldsymbol{y}_t)\boldsymbol{A}' + \boldsymbol{\Sigma}}.$$

- Finally,

$$\begin{aligned}\text{Cov}(\boldsymbol{y}_t, \boldsymbol{y}_{t-\ell}) &= \boldsymbol{A}\,\text{Cov}(\boldsymbol{y}_{t-1}, \boldsymbol{y}_{t-\ell}) = \boldsymbol{A}^2\,\text{Cov}(\boldsymbol{y}_{t-2}, \boldsymbol{y}_{t-\ell}) \\ &= \cdots = \boxed{\boldsymbol{A}^\ell\,\text{Var}(\boldsymbol{y}_t)}.\end{aligned}$$

VAR(1): forecasting

- One-step-ahead forecast:

$$\boldsymbol{y}_{t+1|t} \equiv E[\boldsymbol{y}_{t+1} \mid \{\boldsymbol{y}_s\}_{s\leq t}] = E[\boldsymbol{y}_{t+1} \mid \boldsymbol{y}_t] = \boxed{\boldsymbol{c} + \boldsymbol{A}\boldsymbol{y}_t}.$$

- Multi-step forecast:

$$\begin{aligned}\boldsymbol{y}_{t+h|t} &\equiv E[\boldsymbol{y}_{t+h} \mid \{\boldsymbol{y}_s\}_{s\leq t}] = E[\boldsymbol{y}_{t+h} \mid \boldsymbol{y}_t] \\ &= \sum_{j=0}^{h-1} \boldsymbol{A}^j \boldsymbol{c} + \boldsymbol{A}^h \boldsymbol{y}_t = \boxed{(\boldsymbol{I} - \boldsymbol{A})^{-1}(\boldsymbol{I} - \boldsymbol{A}^h)\boldsymbol{c} + \boldsymbol{A}^h \boldsymbol{y}_t}.\end{aligned}$$

- Exercise: verify these.

VAR(1): estimation

- Since a VAR is just a system of ADL models, we can estimate them in the same way: run an OLS regression *separately* for each of the n equations in the system.

$$y_{it} = \hat{c}_i + \hat{A}_{i1} y_{1,t-1} + \cdots + \hat{A}_{in} y_{n,t-1} + \hat{u}_{it}, \quad i = 1, \ldots, n.$$

- Then stack to get estimators of the parameter vectors/matrices:

$$\hat{\boldsymbol{c}} \equiv \begin{pmatrix} \hat{c}_1 \\ \vdots \\ \hat{c}_n \end{pmatrix}, \quad \hat{\boldsymbol{A}} \equiv \begin{pmatrix} \hat{A}_{11} & \cdots & \hat{A}_{1n} \\ \vdots & \ddots & \vdots \\ \hat{A}_{n1} & \cdots & \hat{A}_{nn} \end{pmatrix}, \quad \hat{\boldsymbol{u}}_t \equiv \begin{pmatrix} \hat{u}_{1t} \\ \vdots \\ \hat{u}_{nt} \end{pmatrix},$$

$$\hat{\boldsymbol{\Sigma}} \equiv \frac{1}{T-1} \sum_{t=2}^{T} \hat{\boldsymbol{u}}_t \hat{\boldsymbol{u}}_t'.$$

VAR(1): properties of OLS estimator

- One can prove that under similar conditions to AR(p) models (including stationarity), the VAR OLS estimator is...
  1. Consistent.
  2. Biased in finite samples.
  3. Asymptotically normal. Standard errors can be computed *as if* we were running OLS regressions on i.i.d. data.

VAR(1): stationarity

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}\boldsymbol{y}_{t-1} + \boldsymbol{u}_t \qquad (\dagger)$$

- When is the VAR(1) model covariance stationary?
- Suppose the square matrix $\boldsymbol{A}$ is diagonalizable:

$$\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1}, \quad \boldsymbol{\Lambda} = \text{diag}(\lambda_1, \ldots, \lambda_n).$$

  Here $\lambda_1, \ldots, \lambda_n$ are the eigenvalues and the columns of $\boldsymbol{V}$ are the corresponding eigenvectors. Assume these are real for now.

- Insert into the VAR equation $(\dagger)$ and pre-multiply by $\boldsymbol{V}^{-1}$ to get

$$\boldsymbol{y}_t^* = \boldsymbol{c}^* + \boldsymbol{\Lambda}\boldsymbol{y}_{t-1}^* + \boldsymbol{u}_t^*, \quad \text{where}$$

$$\boldsymbol{y}_t^* \equiv \boldsymbol{V}^{-1}\boldsymbol{y}_t, \quad \boldsymbol{c}^* \equiv \boldsymbol{V}^{-1}\boldsymbol{c}, \quad \boldsymbol{u}_t^* \equiv \boldsymbol{V}^{-1}\boldsymbol{u}_t.$$

VAR(1): stationarity (cont.)

$$\boldsymbol{y}_t^* = \boldsymbol{c}^* + \boldsymbol{\Lambda}\boldsymbol{y}_{t-1}^* + \boldsymbol{u}_t^*, \quad \text{where} \quad \boldsymbol{y}_t^* \equiv \boldsymbol{V}^{-1}\boldsymbol{y}_t$$

- Recall $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \ldots, \lambda_n)$. Hence, if we write each of the above n equations out explicitly, we get

$$y_{it}^* = c_i^* + \lambda_i y_{i,t-1}^* + u_{it}^*, \quad i = 1, \ldots, n.$$

  So each of the n series y_{it}^* follows a univariate AR(1) process!

- These univariate processes are cov. stationary if and only if $|\lambda_i| < 1$ for all i.

- Clearly, $\boldsymbol{y}_t = \boldsymbol{V}\boldsymbol{y}_t^*$ is cov. stationary if and only if $\boldsymbol{y}_t^*$ is.

- Result: The original VAR(1) model for $\boldsymbol{y}_t$ has a cov. stationary solution if and only if all eigenvalues of $\boldsymbol{A}$ satisfy $|\lambda_i| < 1$.

VAR(1): stationarity (cont.)

- Turns out the previous proof can be extended to the case where $\boldsymbol{A}$ is not diagonalizable, using the "Jordan decomposition".
- Moreover, if some eigenvalues of $\boldsymbol{A}$ are complex numbers, the condition for cov. stationarity remains $|\lambda_i| < 1$.
  - That is, all n eigenvalues must be strictly *inside* the unit circle in the complex plane.
  - Recall: the absolute value of a complex number is sometimes called the **modulus**.

VAR(p) model

- **VAR(p) model**:

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}_1 \boldsymbol{y}_{t-1} + \cdots + \boldsymbol{A}_p \boldsymbol{y}_{t-p} + \boldsymbol{u}_t, \quad E[\boldsymbol{u}_t \mid \{\boldsymbol{y}_s\}_{s<t}] = \boldsymbol{0}.$$

- Write out equation i explicitly:

$$y_{it} = c_i + \sum_{\ell=1}^{p} \sum_{j=1}^{n} A_{\ell,ij} y_{j,t-\ell} + u_{it},$$

  where $A_{\ell,ij}$ is the (i,j) element of $\boldsymbol{A}_\ell$.

- Each variable i depends on p lags of all n variables y_{jt}, $j = 1, \ldots, n$.
- Estimation: separate OLS regression in each of the n equations.
  - Number of covariates in each equation: $n \times p$ (plus one intercept).

VAR: lag order selection

- Could do general-to-specific testing.
- Information criteria:

$$BIC(p) \equiv \log\left(\det\left(\hat{\Sigma}(p)\right)\right) + n(1+np)\frac{\log T_{\text{eff}}}{T_{\text{eff}}},$$

$$AIC(p) \equiv \log\left(\det\left(\hat{\Sigma}(p)\right)\right) + n(1+np)\frac{2}{T_{\text{eff}}},$$

  where $\det(\hat{\Sigma}(p))$ is the determinant of $\hat{\Sigma}(p)$ (the residual var-cov matrix from a VAR regression with p lags).

- Note: a VAR has n equations, each with $1 + np$ coefficients.

Practical coding

- In practice, we don't explicitly run n separate OLS regressions.
- Instead, most statistics programming languages have built-in VAR estimation routines.
- Python: `statsmodels.tsa.VAR`.
  - Can also do lag order selection.
  - Caveat: standard errors computed under the assumption of i.i.d. (and thus homoskedastic) shocks. This is conventional in the literature.

```
# Lag order selection
var_model = sm.tsa.VAR(df[['growth', 'spread']])
lag_order = var_model.select_order(maxlags=12)
print(lag_order.summary())

# Estimation
var_results = var_model.fit(ic='aic')
print(var_results.summary())
```

VAR Order Selection (* highlights the minimums)

	AIC	BIC	FPE	HQIC
0	1.850	1.892	6.362	1.867
1	-0.2777	-0.1517	0.7575	-0.2265
2	-0.5010*	-0.2909*	0.6060*	-0.4156*
3	-0.4799	-0.1857	0.6190	-0.3603
4	-0.4485	-0.07029	0.6388	-0.2948
5	-0.4239	0.03832	0.6549	-0.2361
6	-0.3939	0.1524	0.6751	-0.1719
7	-0.3414	0.2890	0.7120	-0.08523
8	-0.3104	0.4040	0.7350	-0.02005
9	-0.3407	0.4578	0.7137	-0.01623
10	-0.2924	0.5901	0.7499	0.06624
11	-0.3173	0.6493	0.7325	0.07549
12	-0.2991	0.7515	0.7472	0.1278

Results for equation growth

	coefficient	std. error	t-stat	prob
const	0.923150	0.408835	2.258	0.024
L1.growth	0.343830	0.079882	4.304	0.000
L1.spread	-0.258465	0.375531	-0.688	0.491
L2.growth	0.212839	0.077132	2.759	0.006
L2.spread	0.445969	0.378943	1.177	0.239

Results for equation spread

	coefficient	std. error	t-stat	prob
const	0.348306	0.085897	4.055	0.000
L1.growth	-0.010500	0.016783	-0.626	0.532
L1.spread	1.105984	0.078900	14.018	0.000
L2.growth	-0.039451	0.016206	-2.434	0.015
L2.spread	-0.225553	0.079617	-2.833	0.005

Note: growth equation coefficients same as earlier ADL regression.

```
Correlation matrix of residuals
            growth    spread
growth    1.000000  0.036940
spread    0.036940  1.000000
```

Estimate of innovation var-cov matrix:

$$\hat{\boldsymbol{\Sigma}} \equiv \frac{1}{T-p} \sum_{t=p+1}^{T} \hat{\boldsymbol{u}}_t \hat{\boldsymbol{u}}_t',$$

where $\hat{\boldsymbol{u}}_t$ stacks the regression residuals from the n OLS regressions, just like VAR(1) case.

What did we learn?

- Cross-correlogram
- ADL model, Granger causality
- VAR(1) model
  - Stationarity, moments
  - Forecasting
  - Estimation
- VAR(p) model
  - Estimation
  - Lag length selection

Looking ahead

- In the VAR(p) model, how do we...
  - check stationarity?
  - forecast?
  - compute moments?
- Multivariate VARMA model, matrix lag polynomials
- VARs are biased in finite samples. How can we improve the accuracy of our VAR inference when the sample size is small?